

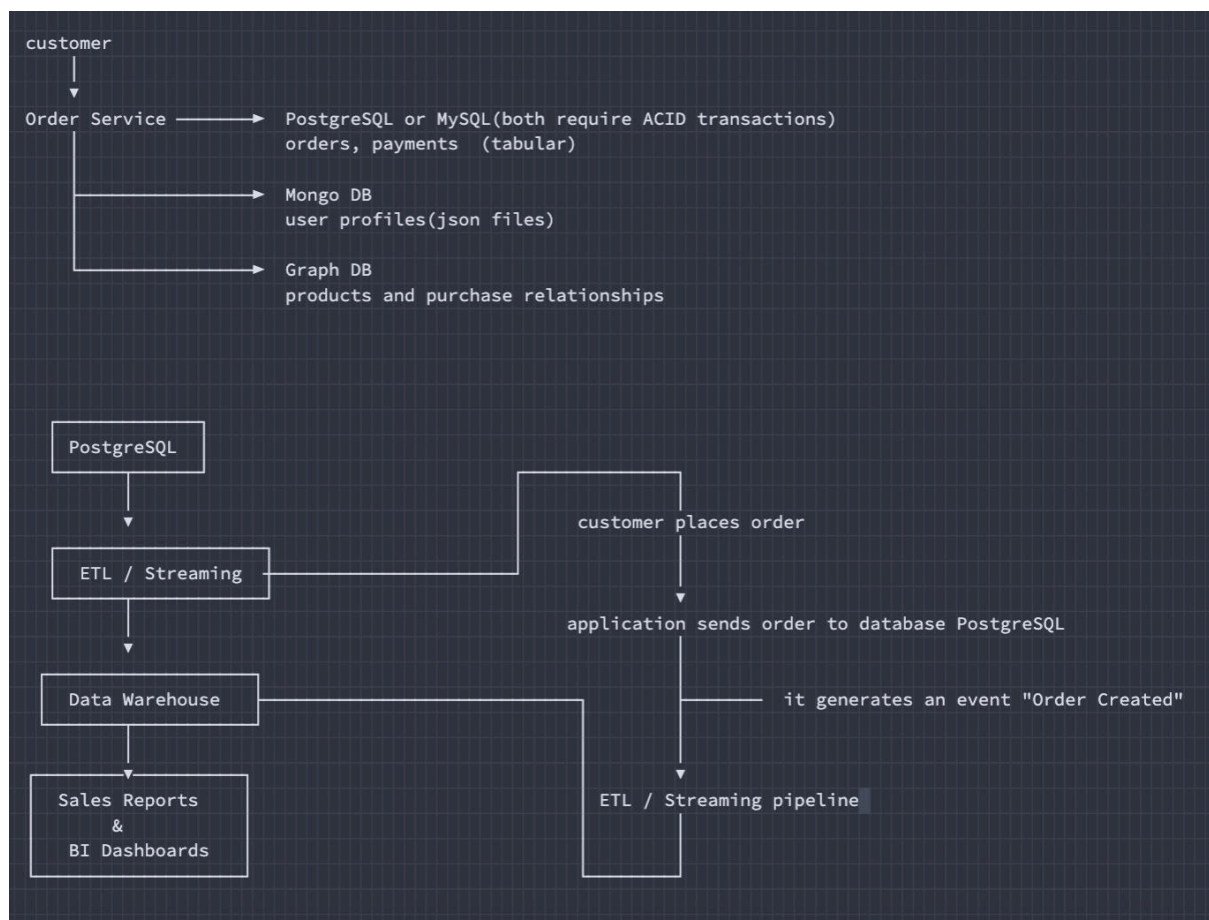
Mini Assignment: **Design storage for an e-commerce platform**

Task: **You are building an e-commerce system that must:**

1. Accept customer orders (must be ACID).
2. Keep user profiles that may add new attributes often.
3. Provide product recommendations based on co-purchases.
4. Produce daily sales analytics.

Deliverables (expected output format)

- A short architecture diagram in text.
- The DB choice for each component with 1-sentence justification.
- One example SQL table definition for the transactional DB.
- One example JSON document for user profile storage.
- Two bullet points describing how data will flow to analytics.



Data Flow: from postgresSQL to analytics(point 3 and 4)

1. after an order is saved in PostgreSQL,
2. an event is created "Order Created"
3. a streaming pipeline listens to these events and send the data to analytics warehouse for near real time dashboards.
4. a scheduled ETL process copies data from PostgreSQL to the analytics db. It cleans and organizes the data for the daily reports while streaming provides updates

> PostgreSQL or MySQL because they are both relational databases that support ACID transactions, therefore a safer choice for an e-commerce platform that will process payment transactions, order management.

> Mongo DB or other document DB for the user profiles because of its flexibility on adding new profile attributes.

> For the products and co-purchase relationships it is better to choose a Graph DB like Neo4j to create relationships between purchases and related products. It can find and suggest products connected to an user's purchase.

//Example to create Orders table in RDBMS

```
CREATE TABLE Orders (  
    order_id INT PRIMARY KEY,  
    customer_id INT NOT NULL,  
    order_date DATE NOT NULL,  
    total_amount DECIMAL(10,2) NOT NULL,  
    status VARCHAR(20) NOT NULL  
);
```

//Example of a JSON file user profile

```
{  
  "user_id": 111,  
  "name": "Klaus Heinz",  
  "email": "k.heinz@example.com",  
  "address": {  
    "street": "Some street"  
    "house_number": 10  
    "city": "Cologne",  
    "country": "Germany"  
  },  
  "preferences": {  
    "language": "German",  
    "favorite_category": "Garden"  
  }  
}
```